Investigating accelerometer accuracy by integrating twice

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1 Introduction

In Vex robotics, there are sensors for quickly measuring locations for accurate repeated robotic automationos. And the most useful one is an IMU. An inertial measurement unit (IMU) is a sensor that measures acceleration at a high rate. Whenever you need to know the precise location of something quickly, double integrating accelerometer data outpaces GPS solutions. They rely on the property that the area under a accelerometer graph can provide enough data used to compute the displacement of the object. In this investigation we will explore numerical and analytical integration of the accuracy of accelerometer data recorded on a phone accelerometer and comparing with video recorded displacement.

2 How to find displacement from acceleration graph

Recall that in kinematics acceleration is defined as the derivative of velocity

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} \tag{1}$$

To find the derivative from first principals, we use limits:

$$\lim_{h \to 0} \frac{v(t+h) - v(t)}{h} \tag{2}$$

and $v(t + h) - v(t)$ essentially means the change in velocity over a short period of time h. And as h approaches 0, it becomes an increasingly more accurate representation of the derivative of acceleration, velocity. And this means that for our data to be accurate, the sample rate has to be high.

Recall the fact that velocity is the integral of acceleration and displacement is the integral of velocity. Integral is the area under a curve. For continuous functions, they are an approximateted by dividing the function into many "rectangles" like shown below where the height of a specific rectangle at a specific point of the function is the y value of the function like shown below.

Figure 1: Function area approximated with rectangles

As the width of the rectangles becomes smaller and smaller, the accuracy increases. So in terms of collecting our data, having data with higher sample rate then it would become more accurate.

But instead of using rectangles to find the area under the curve, we can just use basic geometric equations like the equation for finding the area of a triangle and rectangle.

$$A_{triangle} = \frac{1}{2} b_t \cdot h_t \tag{3}$$

Figure 2: The formula finding the area of a triangle

$$A_{rectangle} = b_r \cdot h_r \tag{4}$$

Figure 3: Formula for the area of a rectangle

the subscripts r and t indicate which base we are refering to (for instance it is the triangles rather than the rectangle's)

Lets say we have two data points, the second data point's y-value could either be higher, lower, or equal to the y-value of the first point (shown below). And the numerical integral is essentially the area under between these two points, which can be calculated with the previous geometric equations illustrated below labeled with the equations

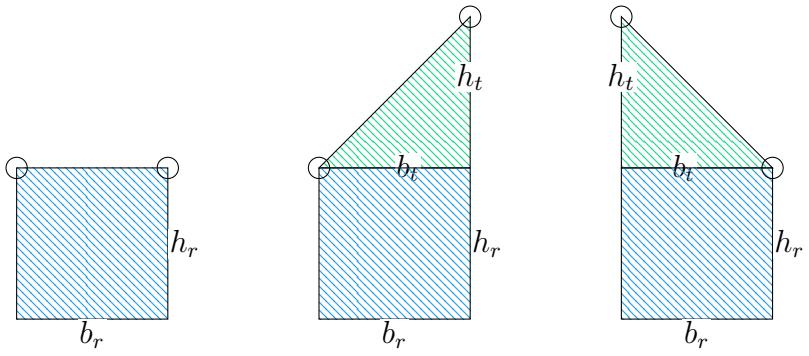


Figure 4: The three conditions of integration

But of course our data isn't just two points, but for the first two points, we find the area, and then the integral function's will use that area as the y-value for the new point. And the point's x-value should be in between the two points that we calculated the area with. And if we have another point, the integral function's corresponding new point's y-value will be the previous point's y-value plus the area between the new point and the previous point, and once again the x-value will be in between the two points in which the area is calculated from. So this is the data we collected from swinging the phone right to left 5 times. Positive means to the e Shown in Figure 5

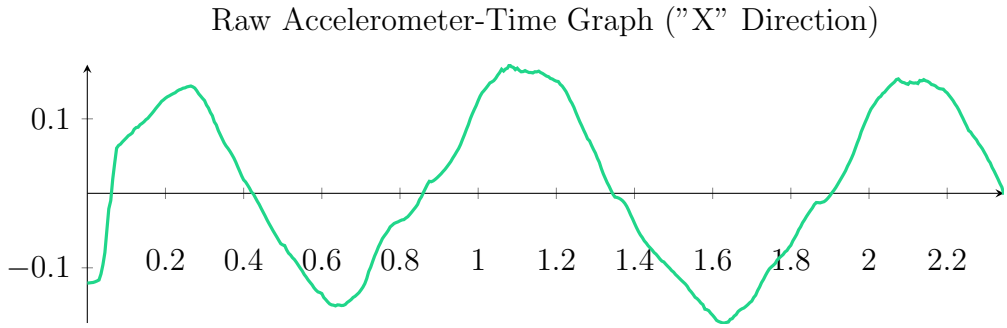


Figure 5: Raw accelerometer data on the X axis

So now have the tools to numerically integrate a piece of data, giving us a velocity-time graph. Shown below in Figure 7

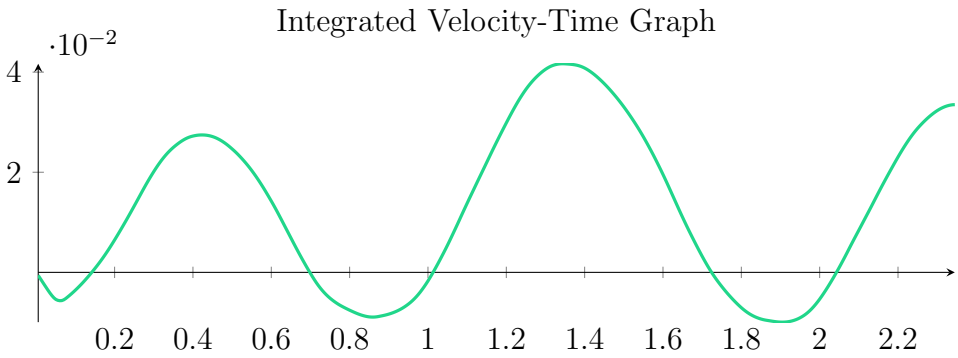


Figure 6: Integrated accelerometer data (velocity)

In theory, if we integrate again, we can finally get a displacement-time graph. But the velocity graph feels a little off because the area is mostly on the upper side of the x-axis, meaning if we were to integrate it, the displacement would start going towards the right rather than what actually happened: going right and left 5 times and then stopping.

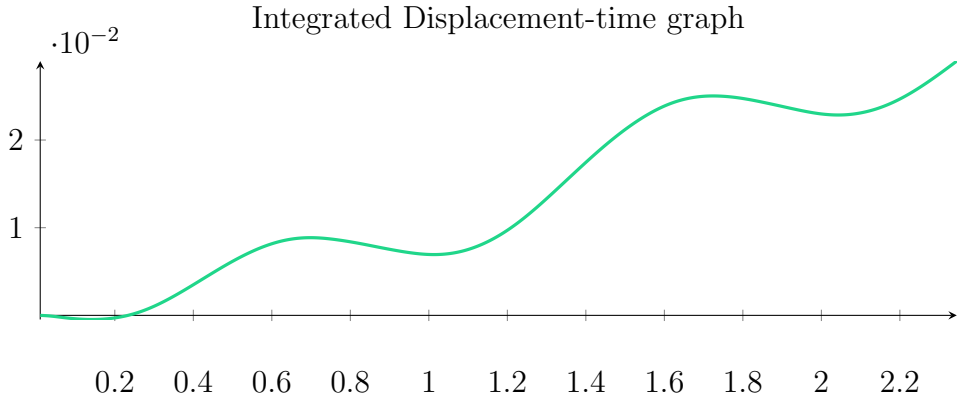


Figure 7: Integrated accelerometer data (Displacement)

One way I thought of to correct this is to center the accelerometer-time graph before integrating it into velocity. As you can see below

Figure 8: Integrating two data points

To calculate the area bounded by the x axis and the line created from two datapoints, we use basic geometry equations

$$\Delta v = \int_a^b a(t)\,\mathrm{d}t = \Delta t \cdot a_0 + \frac{1}{2}\Delta t \cdot a_f \tag{5}$$

2.1 Testing the waters

the same idea will be repeated for each axis and afterwards applied to the calculated velocity values to find displacement. For this first model I moved my phone left and right 5 times on my table starting and ending at 0 velocity.

And integrating the x-axis with the aforementioned algorithm we get this series of integrations expressed graphically (Assume this part will be corrected to the values given by the actual)

Plugging into desmos for Sinusoidal regression we get:

$$y = 0.14272 \cdot \sin\left(6.6486x + 0.216978\right) - 0.00389886 \tag{6}$$

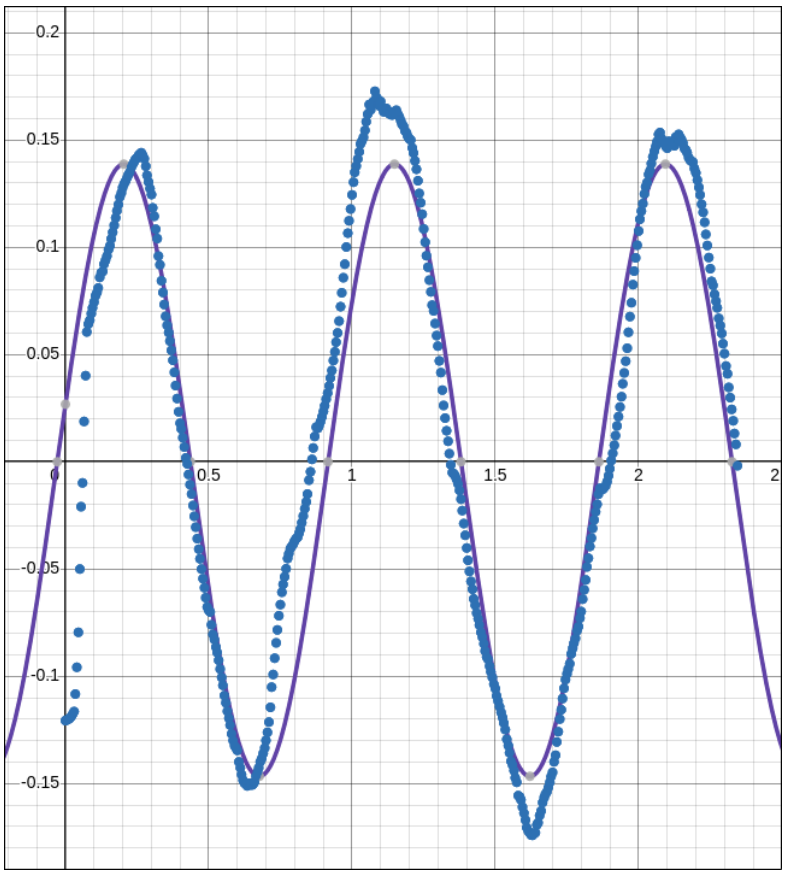


Figure 9: graph of our accelerometer data and the fitted curve

seen in figure 9. We will integrate this analytically twice:

$$v(t) = \int \left(a \cdot \sin\left(bt + c\right) - d\right) \mathrm{d}t \tag{7}$$

where a, b, c, and d are constants. Rearranged, we get

$$v(t) = a \int \sin(bt + c)\, \mathrm{d}t - \int d\, \mathrm{d}t \tag{8}$$

Using u-substitution

$$u = bt + c \tag{9}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = b \tag{10}$$

$$\mathrm{d}t = \frac{\mathrm{d}u}{b} \tag{11}$$

plugging in u and dt:

$$v(t) = a\int \sin(bt + c)\,\mathrm{d}t - \int d\,\mathrm{d}t \tag{12}$$

$$= a\int \sin(u)\frac{1}{b}\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{13}$$

$$= \frac{a}{b}\int \sin(u)\,\mathrm{d}u - \int d\,\mathrm{d}t \tag{14}$$

$$= -\frac{a}{b}\cos(u) - \int d\,\mathrm{d}t \tag{15}$$

$$= -\frac{a}{b}\cos(bt + c) - dt + C_1 \tag{16}$$

and since we know the constant represents the initial velocity, and since we start at rest, we need to find the constant C_1 such that $v(0) = 0$

$$v(0) = -\frac{a}{b}\cos((0)\,b + c) - (0)\,d + C_1 = 0 \tag{17}$$

$$\frac{a}{b}\cos(c) = C_1 = 0.0209628448456 \tag{18}$$

$$\tag{19}$$

For now, we will keep in mind that $C_1 = 0.0209628448456$ but continue using C_1 throughout integration for convenience. To find the displacement function with respect to time we integrate again.

$$s(t) = \int \left(-\frac{a}{b}\cos(bt + c) - dt + C_1\right)\mathrm{d}t \tag{20}$$

$$s(t) = -\frac{a}{b}\int \cos(bt + c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{21}$$

$$s(t) = -\frac{a}{b}\int \cos(bt + c)\,\mathrm{d}t - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{22}$$

$$\tag{23}$$

Using the same u-substitution:

$$s(t) = -\frac{a}{b}\int \cos(u)\frac{\mathrm{d}u}{b} - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t \tag{24}$$

$$s(t) = -\frac{a}{b^2}\sin(u) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{25}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \int dt\,\mathrm{d}t + \int C_1\,\mathrm{d}t + C_2 \tag{26}$$

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{27}$$

$$\tag{28}$$

and since this data is filmed as relative displacement from original point (seen as 0) we need to find the value of C_2 such that $s(0) = 0$. [Ai said wording was bad, I don't understand why]

$$s(0) = -\frac{a}{b^2}\sin(b\,(0)+c) - \frac{d\,(0)^2}{2} + C_1\,(0) + C_2 = 0 \tag{29}$$

$$0 = -\frac{a}{b^2}\sin(b\,(0)+c) + C_2 \tag{30}$$

$$\frac{a}{b^2}\sin(b\,(0)+c) = C_2 = 0.000695067592524 \tag{31}$$

$$\tag{32}$$

Again we will keep $C_2 = 0.000695067592524$ in mind. Resulting in a final function $s(t)$:

$$s(t) = -\frac{a}{b^2}\sin(bt+c) - \frac{dt^2}{2} + C_1 t + C_2 \tag{33}$$

drawing a graph of progressive integration is seen in figure 10

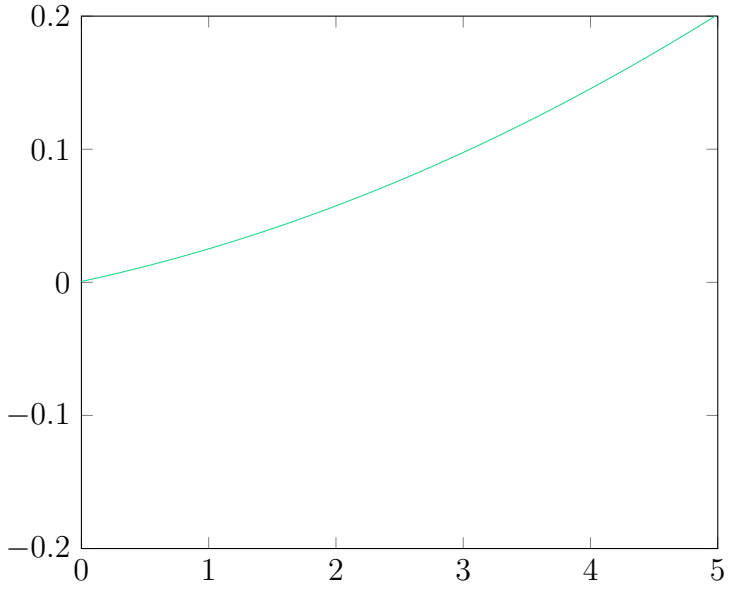


Figure 10: graph of the model being integrated twice

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We have a displacement/time graph from video. However it is a different sampling rate (120hz) than the recorded datapoints (100hz). Hence we will attempt to align by drawing a line between the points in the recorded accelerometer data, and interpolate the value.

3 Appendix

3.1 Python code

```
dt = final_time - initial_time
dx_accel = final_accel - initial_accel
dx_vel = (dt * initial_accel) + ((1/2) * dt * dx_accel)
# triangle area + square area = area under accel graph = change in velocity
```

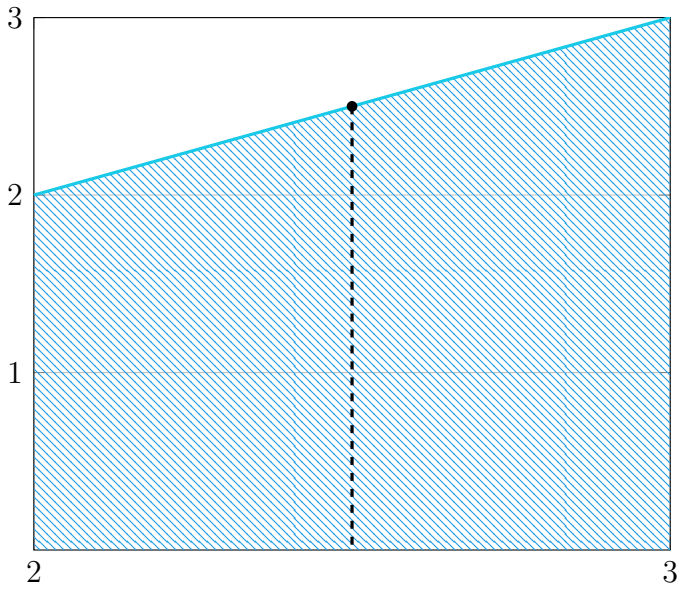


Figure 11: Illustration of interpolating values from two datapoints $x = 2, 3$